

HA'APAI, Tonga

WMO: 917840

Lat: 19.777S

Lon: 174.341W

Elev: 31

StdP: 14.68

Time Zone: 13.00 (E13)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	66.6	68.2	54.9	64.2	70.8	56.5	68.2	71.4	23.5	73.9	22.4	73.3	4.4	180	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.0	86.4	80.1	86.0	79.8	85.3	79.2	81.7	84.9	80.9	84.3	80.2	83.6	8.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	159.9	84.4	79.9	155.6	83.7	79.1	151.4	82.9	45.6	85.3	44.7	84.7	44.0	84.3	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	20.0	18.4	DB	63.9	88.6	1.3	1.0	62.9	89.4	62.1	90.0	61.4	90.6	60.4	91.3	(4)
(5)			WB	58.2	83.6	1.4	0.8	57.3	84.2	56.5	84.6	55.7	85.1	54.7	85.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	77.9	80.8	81.4	81.2	79.8	77.7	76.0	74.6	74.2	74.6	76.4	78.3	80.0
	DBStd	3.35	1.92	1.82	1.64	1.80	2.02	2.43	2.34	2.27	2.75	2.33	2.18	1.88
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10179	955	878	966	893	859	780	762	752	738	818	848	930
	CDD65	4704	490	458	501	443	394	330	297	287	288	353	398	465
	CDH74	34934	4935	4772	5149	3976	2599	1599	990	827	1087	1875	2926	4199
	CDH80	5466	1102	1182	1161	591	163	61	21	20	32	103	312	718
Wind	WSAvg	9.4	9.1	8.5	8.0	9.2	10.1	9.2	9.3	9.9	9.3	10.0	9.6	10.0
Precipitation	PrecAvg	66.1	6.7	7.6	10.0	6.9	4.2	2.9	3.6	4.0	4.1	3.8	4.8	6.0
	PrecMax	104.8	17.4	19.1	23.3	19.6	16.9	9.1	9.1	13.4	11.6	13.3	20.1	24.1
	PrecMin	32.5	0.4	2.0	2.0	0.1	0.0	0.3	0.1	0.5	0.1	0.2	0.1	0.3
	PrecStd	17.4	4.3	4.3	5.2	4.6	3.4	2.1	2.5	2.9	3.0	3.1	4.4	4.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.4	88.1	87.6	86.1	84.3	82.7	82.3	82.1	82.4	83.1	85.1	86.4
		MCWB	80.9	81.4	81.3	79.2	78.3	77.1	76.6	76.2	76.5	76.9	78.3	78.8
	2%	DB	86.2	86.4	86.2	84.8	82.7	81.3	80.6	80.5	80.9	82.3	83.9	85.4
		MCWB	80.3	80.3	80.1	78.2	76.8	76.2	75.3	75.2	75.1	76.2	77.4	78.4
	5%	DB	85.5	86.0	85.6	84.1	82.0	80.6	79.2	79.1	79.8	81.0	82.7	84.3
		MCWB	79.8	79.9	79.6	77.7	76.2	75.6	73.9	73.9	74.1	75.3	76.3	78.0
	10%	DB	84.3	84.9	84.5	82.9	80.8	79.4	78.4	77.8	78.7	80.3	82.0	83.5
		MCWB	78.9	79.2	78.7	76.9	75.2	74.6	73.1	72.7	73.1	74.8	75.8	77.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.1	83.2	82.4	81.0	79.8	79.2	78.3	77.8	78.1	79.1	79.9	81.2
		MCDB	85.2	86.4	85.8	83.9	82.2	81.1	80.9	80.1	80.8	81.4	83.2	84.3
	2%	WB	81.3	81.8	81.3	79.9	78.6	78.0	76.9	76.5	76.7	77.6	78.8	80.0
		MCDB	84.5	85.1	84.7	83.1	81.0	80.1	79.2	79.2	79.6	80.4	82.1	83.3
	5%	WB	80.5	80.9	80.5	79.1	77.8	77.1	75.6	75.3	75.6	76.6	78.0	79.2
		MCDB	83.8	84.4	84.0	82.4	80.4	79.3	78.0	77.8	78.3	79.7	81.4	82.6
	10%	WB	79.8	80.1	79.8	78.4	77.0	76.2	74.5	74.2	74.6	75.8	77.1	78.6
		MCDB	83.1	83.7	83.2	81.8	79.9	78.6	77.1	77.0	77.5	78.9	80.7	82.1
Mean Daily Temperature Range	5% DB	MDBR	6.8	7.0	6.7	6.3	5.8	6.1	6.4	6.3	6.8	6.7	6.5	6.7
		MCDBR	7.6	7.8	7.6	7.4	6.4	6.9	7.2	7.3	7.3	7.1	7.2	7.4
		MCWBR	4.5	4.4	4.4	4.5	4.0	4.7	5.1	4.8	4.7	4.5	4.3	4.2
	5% WB	MCDBR	7.0	7.4	6.9	6.6	6.0	6.3	6.4	6.7	6.8	6.6	6.9	6.9
		MCWBR	4.5	4.5	4.6	4.6	4.0	5.0	5.8	4.9	5.2	4.8	4.8	4.6
Clear-Sky Solar Irradiance	taub	0.387	0.380	0.378	0.396	0.380	0.377	0.366	0.371	0.396	0.379	0.380	0.385	
	taud	2.613	2.641	2.656	2.583	2.610	2.598	2.622	2.603	2.528	2.576	2.600	2.595	
	Ebn at Noon	304	302	296	278	271	265	272	281	285	299	304	304	
	Edh at Noon	33	31	30	30	27	27	27	29	33	33	33	33	
All-Sky Solar Radiation	RadAvg	2009	1930	1788	1491	1284	1158	1246	1430	1667	1901	2003	2077	
	RadStd	194	192	132	170	81	90	75	124	125	134	159	142	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	+1.87	N/A	+1.28	+1.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page